PROJECT TREE - IFM AL1403 SENSOR NETWORK

Ten IFS240 sensors are shown once on the detailed AL100, AL101 and AL103 port-map pages.

PLC_1 - S7-1512C-1 PN

- **AL100** — 4 IFS240
- **AL101** — 3 IFS240
- **AL103** — 3 IFS240

TOTAL: 10 IFS240 INSTALLED - 0 SPARE - 0 DUPLICATE

See the three detailed port maps for sensor function, M12 connection and configuration status.

HANDHELD PENDANT

Pendant connected | JOG enable | Machine rotation JOG | UP | DOWN

Direct 24 VDC PLC inputs - physical addresses to be confirmed

HANDHELD PENDANT / JOG - FUNCTIONAL MAP

Dedicated operator-control page. Physical PLC addresses and contact allocation remain to be confirmed from the final electrical design.



OMRON A4EG-RM2R041

PLC_1 - S7-1512C-1 PN

PENDANT INTERFACE

24 VDC discrete control interface - addresses TBC

Pendant Connected

Connection/plug presence proof

Enable / consent

Enabling-switch permissive; does not bypass Safety

JOG machine rotation

Momentary JOG request to main drive

Pendant UP

Momentary filler/capper lift UP request

Pendant DOWN

Momentary filler/capper lift DOWN request

PLC PERMISSIVES

- Manual mode active
- Safety OK
- Pendant connected
- JOG enable/consent active
- No lift movement active
- Main drive ready

OUTPUT BEHAVIOUR

JOG rotation is momentary.

Release command -> motor request OFF.

Maximum JOG frequency: 15 Hz.

UP and DOWN are mutually exclusive.

HOLD POINT: confirm contact blocks, cable cores, connector pinout and PLC input addresses before cabinet wiring or download.

SENSORS - ENLARGED M12 ELECTRICAL CONNECTIONS

Sensor and connector are separated; pin numbers are enlarged for workshop wiring.

PHOTOELECTRIC SENSORS

IFM O6P301

M12 A-coded - physical connector shown at right





PIN 1 = L+ / +24 VDC
PIN 3 = L- / 0 VDC
PIN 4 = PNP DIGITAL OUTPUT

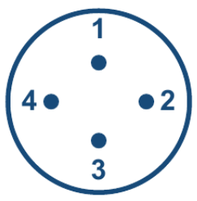
Pin 2 is not assigned in this mapping.

INDUCTIVE SENSORS

IFM IFS240 (10 units)

M12 A-coded - physical connector shown at right





PIN 1 = L+ / +24 VDC
PIN 3 = L- / 0 VDC
PIN 4 = SIO DIGITAL INPUT

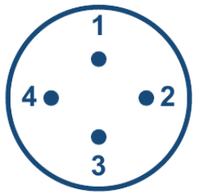
Pin 2 is not assigned in this mapping.

PRODUCT / WATER PRESENCE

IFM LMT100

M12 A-coded - physical connector shown at right





PIN 1 = L+ / +24 VDC
PIN 3 = L- / 0 VDC
PIN 4 = Q/C - VERIFY MODE

SIO/IO-Link mode must be confirmed in the final configuration.

MACHINE AIR PRESSURE

Sensor model to be confirmed

M12 A-coded - physical connector shown at right





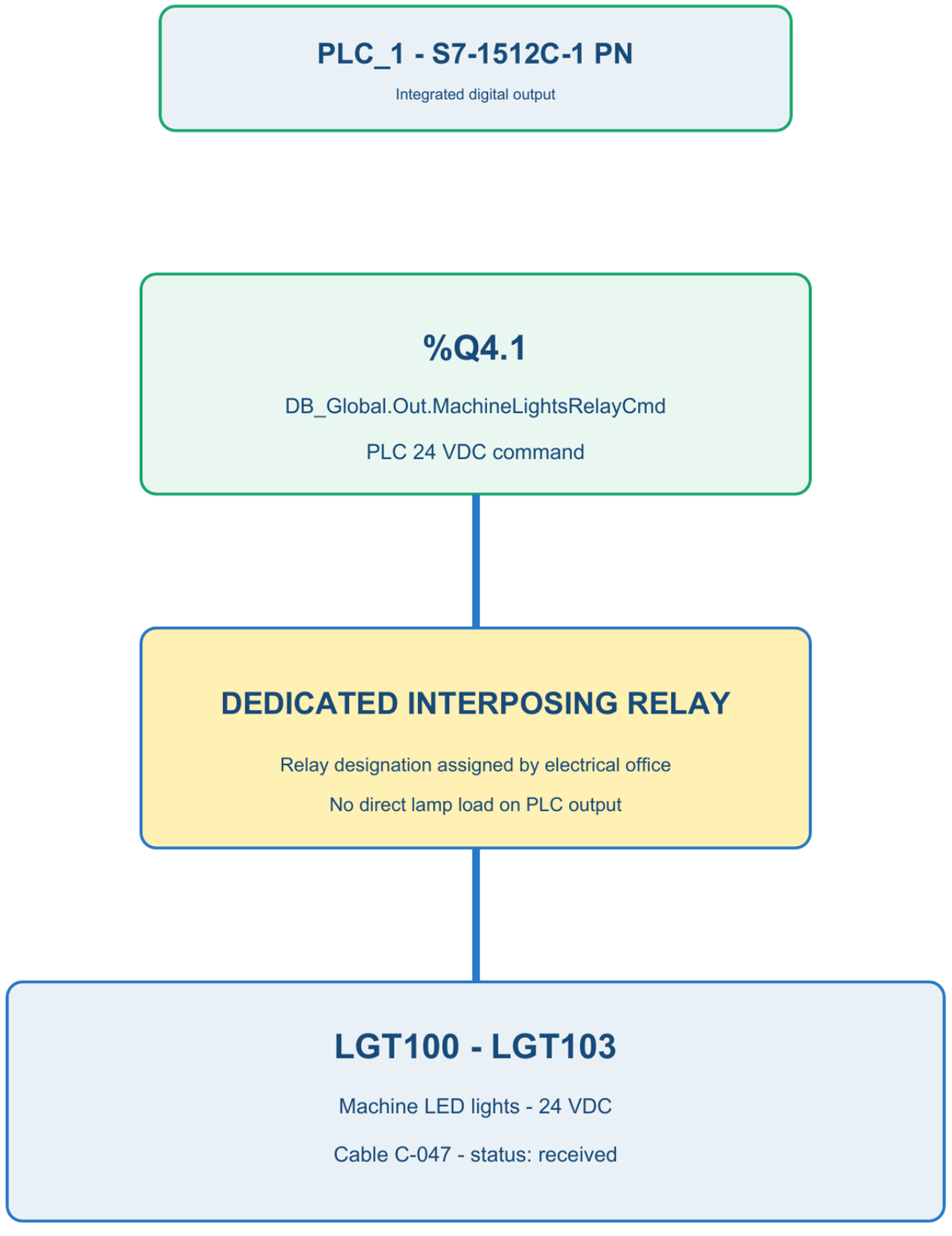
PIN 1 = L+ / +24 VDC
PIN 3 = L- / 0 VDC
PIN 4 = NOT DEFINED

Do not wire until the model and datasheet are approved.

NOTE: no undocumented address or pin has been invented; always verify the purchased device datasheet.

MACHINE LIGHTS 24 VDC - PLC DIRECT OUTPUT

Dedicated machine-light command page; this circuit is separate from the IO-Link network.



ROUTE: PLC %Q4.1 -> interposing relay -> LGT100-LGT103

HOOD / VENTILATION - PLC COMMAND AND FEEDBACK

Dedicated hood page; confirmed points and unresolved points are shown separately.



OPEN POINT - PSH100

Definitive PLC baseline: hood/filter alarm mapped to %I12.4.

Cable schedule: PSH100 listed as IO-Link / SIO on an AL1403 port (planned).

Resolve before cabinet construction; no AL1403 port is assigned in this document.

CURRENT ROUTE: PLC %Q4.2 -> relay/contactor -> filter/ventilation unit; feedbacks return to %I12.4, %I12.6 and %I12.7

AL100 - IFM AL1403 GRAPHICAL PORT MAP

PROFINET input %IB234; output %QB135. Every line below is one physical port.

PLC_1 - S7-1512C-1 PN

PROFINET controller

AL100

IFM AL1403 - 8 physical ports P1-P8

P1 - INPUT - Bottle Shortage 1

IFM O6P301

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED



PHOTOELECTRIC SENSOR

1=L+ 3=L- 4=Q/C

P2 - INPUT - Bottle Shortage 2

IFM O6P301

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED



PHOTOELECTRIC SENSOR

1=L+ 3=L- 4=Q/C

P3 - INPUT - Outfeed Accumulation 1

IFM O6P301

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED



PHOTOELECTRIC SENSOR

1=L+ 3=L- 4=Q/C

P4 - INPUT - Outfeed Accumulation 2

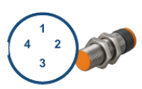
IFM O6P301

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED



PHOTOELECTRIC SENSOR

1=L+ 3=L- 4=Q/C



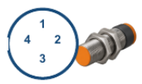
PROXIMITY SENSOR

1=L+ 3=L- 4=Q/C

P5 - INPUT - Capper cylinder presence/position

IFS240-01

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED



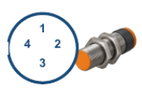
PROXIMITY SENSOR

1=L+ 3=L- 4=Q/C

P6 - INPUT - Cap distributor rotation

IFS240-02

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED



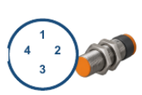
PROXIMITY SENSOR

1=L+ 3=L- 4=Q/C

P7 - INPUT - Capper maximum height

IFS240-03

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED



PROXIMITY SENSOR

1=L+ 3=L- 4=Q/C

P8 - INPUT - Capper minimum height

IFS240-04

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED

IFS240 COUNT ON AL100: 4 | Total across AL100 + AL101 + AL103 = 10.

AL101 - IFM AL1403 GRAPHICAL PORT MAP

PROFINET input %IB198; output %QB134. Every line below is one physical port.

PLC_1 - S7-1512C-1 PN

PROFINET controller

AL101

IFM AL1403 - 8 physical ports P1-P8

P1 - INPUT - Cap Present Pick and Place

IFM O6P301

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED



PHOTOELECTRIC SENSOR

1=L+ 3=L- 4=Q/C

P2 - INPUT - Cap Hopper Low Level

IFM O6P301

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED



PHOTOELECTRIC SENSOR

1=L+ 3=L- 4=Q/C

P3 - INPUT - Cap Channel Request

IFM O6P301

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED



PHOTOELECTRIC SENSOR

1=L+ 3=L- 4=Q/C

P4 - INPUT - Cap Channel Empty

IFM O6P301

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED



PHOTOELECTRIC SENSOR

1=L+ 3=L- 4=Q/C

P5 - INPUT - Machine Air Pressure OK

Pressure sensor - model TBC

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED



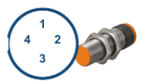
PRESSURE SENSOR

1=L+ 3=L- 4=Q/C

P6 - INPUT - Machine zero / filler valve no. 1

IFS240-05

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED



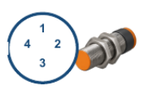
PROXIMITY SENSOR

1=L+ 3=L- 4=Q/C

P7 - INPUT - Filler cylinder presence/position

IFS240-06

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED



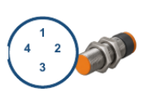
PROXIMITY SENSOR

1=L+ 3=L- 4=Q/C

P8 - INPUT - Filler anti-rotation

IFS240-07

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED



PROXIMITY SENSOR

1=L+ 3=L- 4=Q/C

IFS240 COUNT ON AL101: 3 | Total across AL100 + AL101 + AL103 = 10.

AL102 - IFM AL1403 GRAPHICAL PORT MAP

PROFINET input area begins %IB162; output byte %QB133. Every line below is one physical port.

PLC_1 - S7-1512C-1 PN
PROFINET controller

SCALANCE XB116
PROFINET / Ethernet switch

Any available RJ45 port

AL102
IFM AL1403 - 8 physical ports P1-P8

P1 - INPUT - Product Pressure
PT-PRODUCT | IFM PN7094 candidate | IO-Link | ai_ProductPressure | ASSIGN DEVICE AT BUILD / NOT CONFIGURED



PRESSURE SENSOR
1=L+ 3=L- 4=C/Q

P2 - INPUT - Pneumatic/Process Pressure Diagnostic
PT-AIR | IFM PN7094 candidate | IO-Link | ai_AirPressure | ASSIGN DEVICE AT BUILD / NOT CONFIGURED



PRESSURE SENSOR
1=L+ 3=L- 4=C/Q

P3 - INPUT - CIP/Service Pressure Diagnostic
PT-CIP | IFM PN7094 candidate | IO-Link | ai_CIPPressure | ASSIGN DEVICE AT BUILD / NOT CONFIGURED



PRESSURE SENSOR
1=L+ 3=L- 4=C/Q

P4 - INPUT - Spare Pressure Measurement
PT-SPARE | IFM PN7094 candidate | IO-Link | ai_SparePressure | SPARE INSTRUMENT / NOT CONFIGURED



PRESSURE SENSOR
1=L+ 3=L- 4=C/Q

P5 - INPUT - Product Level/Continuous Level
ODUCT | IFM LR2050 candidate | IO-Link | ai_ProductLevel | ASSIGN DEVICE AT BUILD / NOT CONFIGURED



LEVEL SENSOR
1=L+ 3=L- 4=C/Q

P6 - INPUT - CIP/Process Level
CIP | IFM LR2050 candidate | IO-Link | ai_CIPLevel | ASSIGN DEVICE AT BUILD / NOT CONFIGURED



LEVEL SENSOR
1=L+ 3=L- 4=C/Q

P7 - INPUT - Process Temperature
ROCESS | IFM TA2405 candidate | IO-Link | ai_ProcessTemperature | ASSIGN DEVICE AT BUILD / NOT CONFIGURED



TEMP SENSOR
1=L+ 3=L- 4=C/Q

P8 - OUTPUT - Stack Light Output Hub
DO-HUB-TL100 | IFM AL2330 | IO-Link 8I/8O
%QB156..%QB163 | GREEN / ORANGE / RED / BUZZER

AL103 - IFM AL1403 GRAPHICAL PORT MAP

PROFINET input %IB270; output %QB136. Every line below is one physical port.

PLC_1 - S7-1512C-1 PN

PROFINET controller

AL103

IFM AL1403 - 8 physical ports P1-P8

P1 - INPUT - Product/water presence before pump

IFM LMT100

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED



LEVEL SENSOR

1=L+ 3=L- 4=Q/C

P2 - INPUT - Filler maximum height

IFS240-08

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED



PROXIMITY SENSOR

1=L+ 3=L- 4=Q/C

P3 - INPUT - Filler minimum height

IFS240-09

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED



PROXIMITY SENSOR

1=L+ 3=L- 4=Q/C

P4 - INPUT - Metal dummy bottle presence for washing

IFS240-10

M12: 1=L+ | 3=L- | 4=PNP output | address NOT CONFIGURED



PROXIMITY SENSOR

1=L+ 3=L- 4=Q/C

P5 - SPARE - Future smart sensor

SPARE

NO DEVICE INSTALLED

P6 - SPARE - Future smart sensor

SPARE

NO DEVICE INSTALLED

P7 - SPARE - Future smart sensor

SPARE

NO DEVICE INSTALLED

P8 - SPARE - Future smart sensor

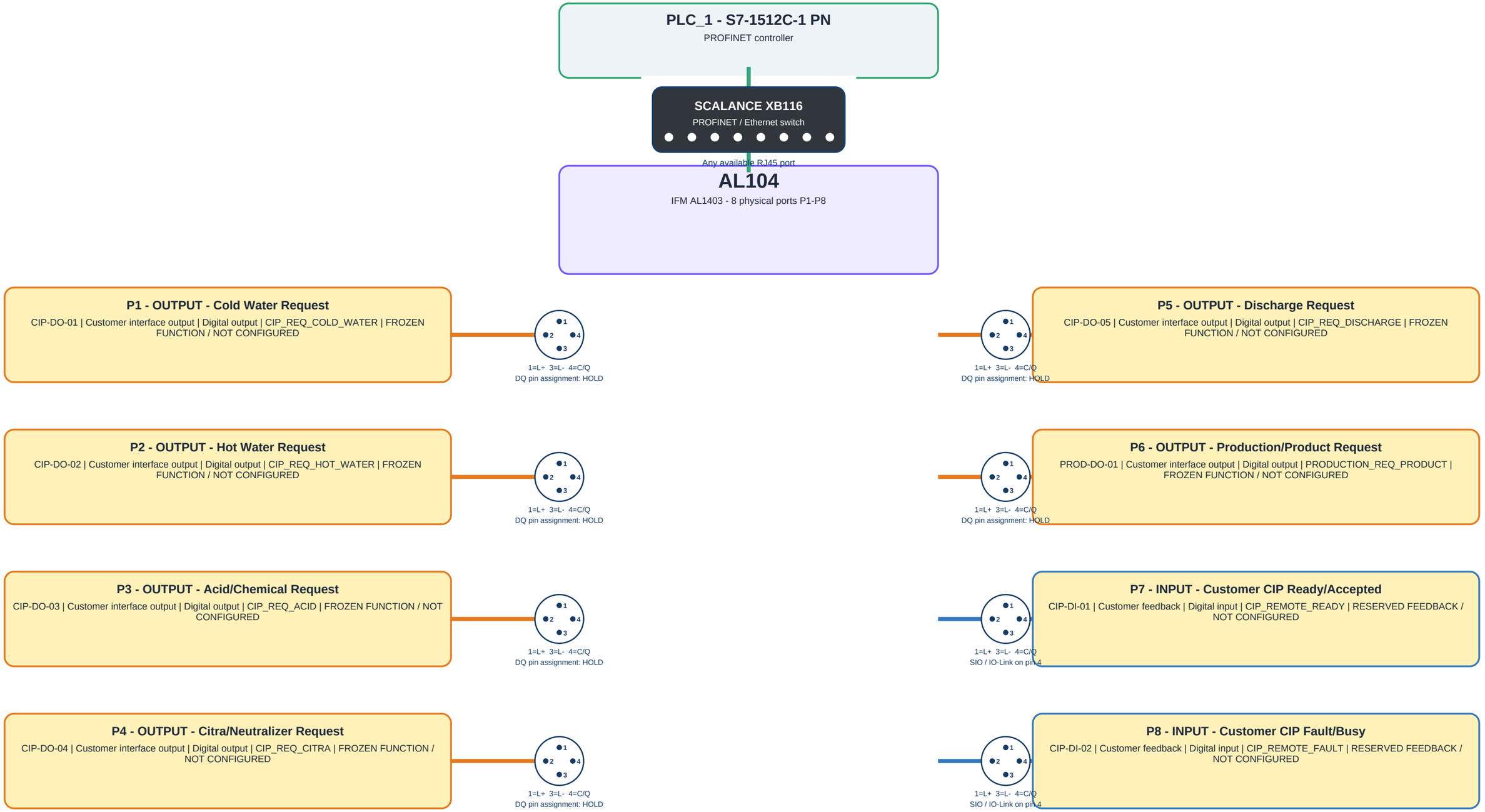
SPARE

NO DEVICE INSTALLED

IFS240 COUNT ON AL103: 3 | Total across AL100 + AL101 + AL103 = 10.

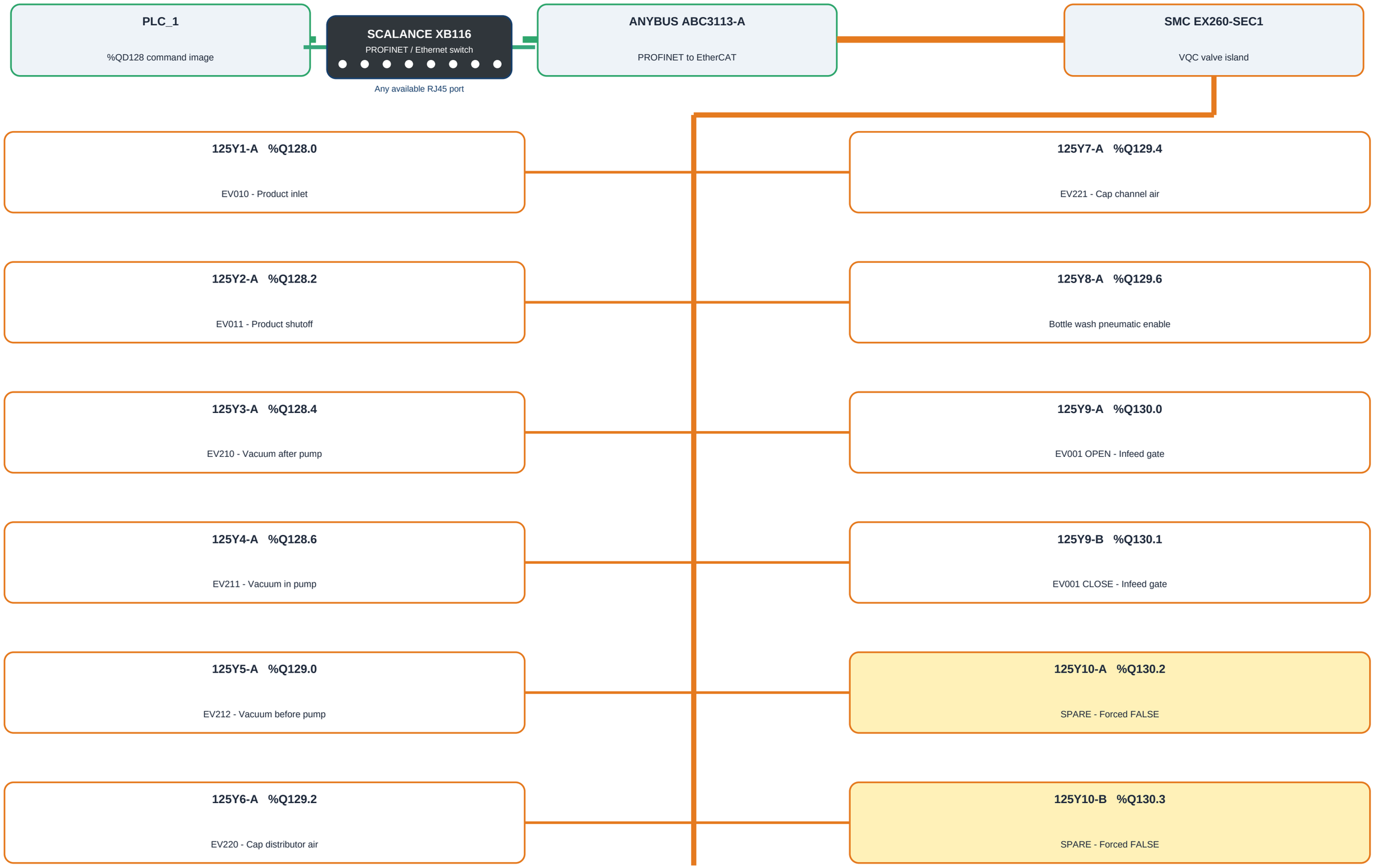
AL104 - IFM AL1403 GRAPHICAL PORT MAP

PROFINET input area begins %IB126; output byte %QB132. Every line below is one physical port.



ANYBUS - SMC EX260 CLEAN GRAPHICAL I/O MAP

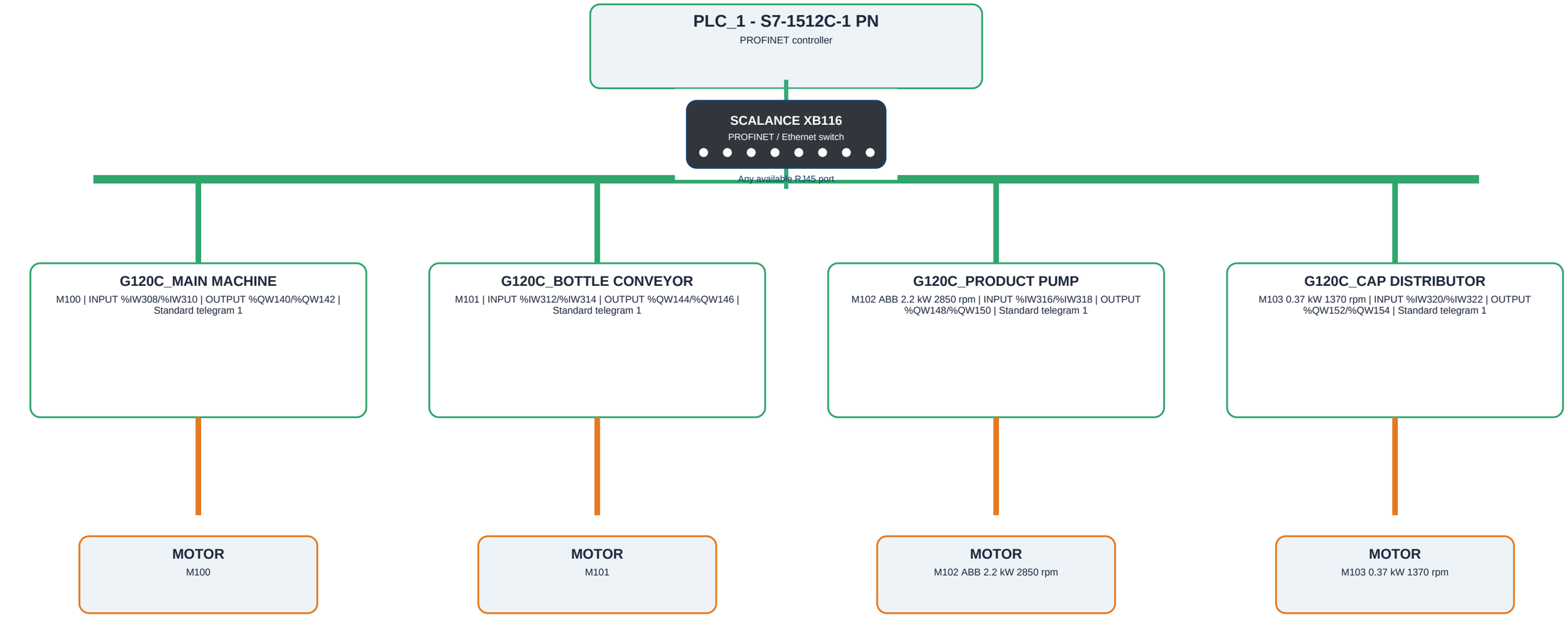
PLC PROFINET -> HMS ABC3113-A -> EtherCAT -> SMC EX260-SEC1 | output image %QD128



Feedback: %I11.2 GateOpenFB | %I11.3 GateClosedFB | AnybusGatewayOK from PROFINET diagnostics | independent EtherCAT feedback not configured

FOUR SINAMICS G120C - GRAPHICAL NETWORK MAP

Standard telegram 1: status word + actual speed to PLC; control word + speed setpoint from PLC



Before production: verify target identity, motor plate, command/setpoint source, safety state and rotation direction.

E100 - IFM RO3101 GRAPHICAL CONNECTION MAP

Same AL102 visual standard. One physical encoder, one counting branch and one safety-monitoring branch. No photographs.

PLC_1 - S7-1512C-1 PN

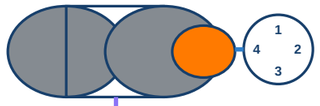
Controller

TM COUNT 2x24V - 6ES7550-1AA01-0AB0

SIMATIC S7-1500 technology module - primary A/B/Z counting interface

S7-1500 FRONT CONNECTOR

6ES7592-1BM00-0XB0
40-pole push-in connector
For 35 mm wide S7-1500 modules



IFM RO3101

E100 - IFM RO3101 / C-001 / HTL A-B-Z / 2048 PPR

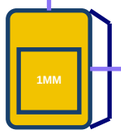
ENCODER POWER AND SIGNALS

+24 VDC / 0 V
HTL channels A, B and Z
Shielded cable; segregate from motor power

PLC DATA

EncoderCount - TM Count value
EncoderHealthy - %I10.2 baseline
Direction, Z-reference and scaling test

Approved encoder adapter / signal tap



PNOZ m EF 1MM - 772170

Safe speed / standstill monitoring

PIN NUMBERS: USE THE OFFICIAL RO3101 CABLE DRAWING AND THE APPROVED SIEMENS/PILZ ADAPTER DRAWING. DO NOT CONNECT TO AL100-AL104.

SAFE1 - PILZ PNOZMULTI 2 SAFETY ARCHITECTURE

Hierarchical project tree. Final quantity: 11 guard doors DS100-DS110. Vector/isometric drawings only.

PLC_1 - S7-1512C-1 PN

Non-safety diagnostics only

SCALANCE XB116

PROFINET / Ethernet switch

PNOZMULTI CHIPCARD

Pilz 779211 - 32 kB
Stores the approved PNOZmulti safety project
Accessory of PNOZ m B0

PNOZ m ES PROFINET - 772138

Diagnostic communication module

PNOZ m B0 - 772100

Safety controller / PNOZmulti authority



PNOZ m EF 16DI

772140

SAFETY INPUTS

11 x PSEN me1S/1AR - 570001
E-stops and safety feedbacks
Final channel allocation in Pilz project



PNOZ m EF 8DI4DO

772142

SAFETY OUTPUTS

Drive STO and safety contactors
Pneumatic dump / EDM
Restart prevention



PNOZ m EF 1MM

772170

SAFE MOTION

E100 RO3101 encoder branch
Safe speed / standstill
Approved adapter topology

PNOZ y2 3NO

2A000014

RELAY OUTPUT

PNOZ y2 3NO
3 safety NO contacts
Approved load circuit

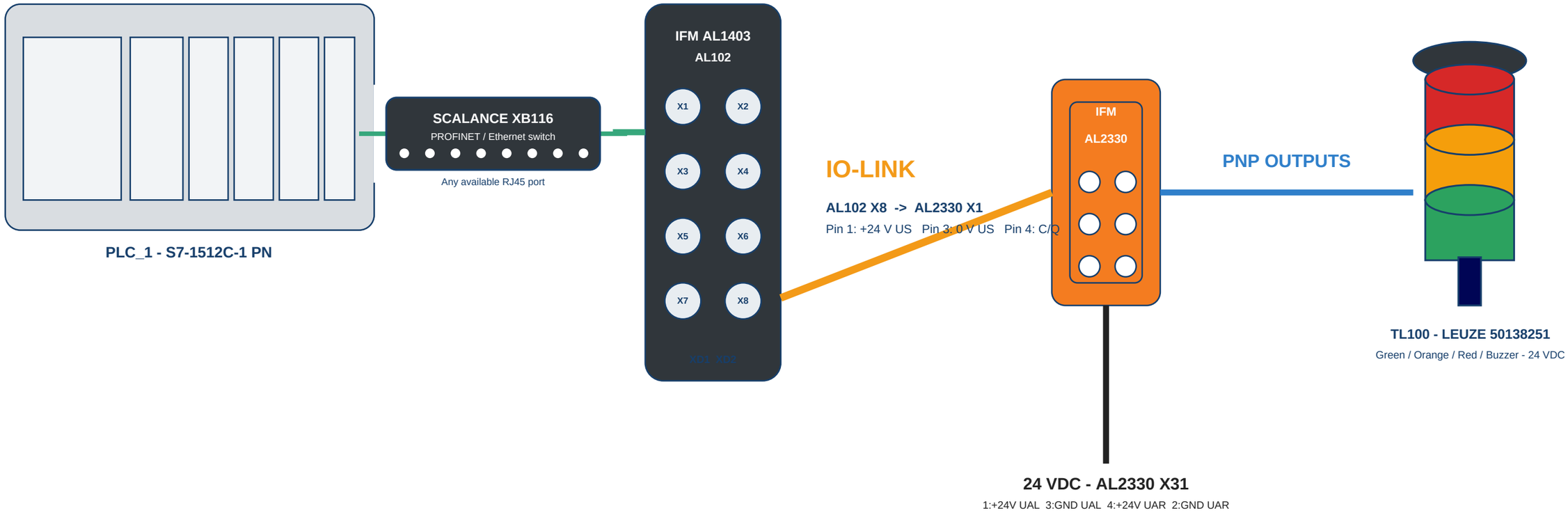
11 GUARD-DOOR INTERLOCKS - PSEN me1S/1AR - 24 V AC/DC - 2 NO + 2 NC - SPRING LOCKING - 1500 N



PILZ IS THE SAFETY AUTHORITY. FINAL TERMINALS AND CHANNELS REQUIRE THE APPROVED RISK ASSESSMENT, ELECTRICAL SAFETY DRAWING AND PNOZMULTI PROJECT.

TL100 - LEUZE STACK LIGHT CONNECTION

Graphical architecture - physical ports and electrical connections.



OUTPUT WIRING

X1.0 pin 4 - DO1
GREEN
Machine running / normal

X1.0 pin 2 - DO2
ORANGE
Warning / attention

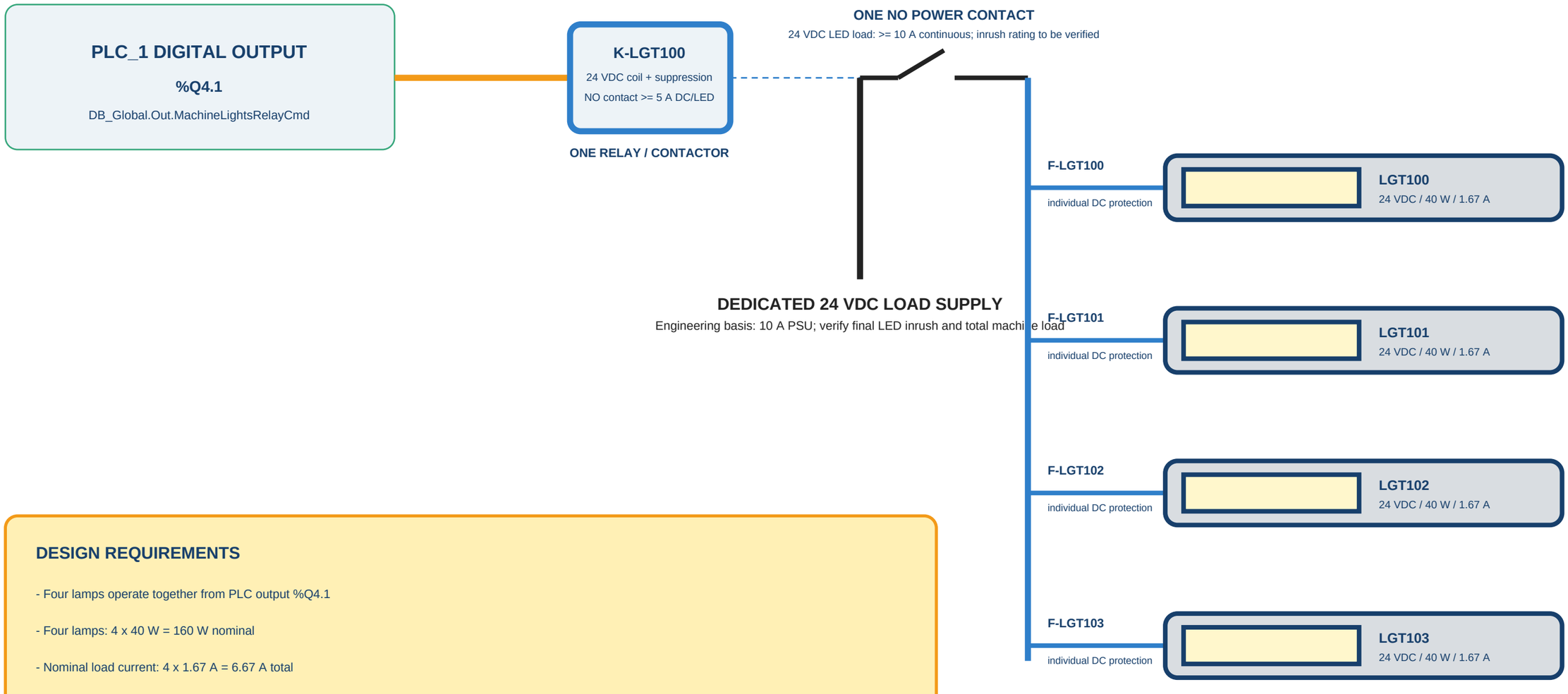
X1.1 pin 4 - DO1
RED
Fault / stop

X1.1 pin 2 - DO2
BUZZER
Acoustic alarm

Every used AL2330 output socket: pin 3 = GND UA. X1.2-X1.5 remain available. Field cable C-046; final conductor colors per Leuze instructions.

LGT100-LGT103 - FOUR MACHINE WORK LIGHTS

One PLC output commands one 24 VDC relay/contactor. A single power contact switches all four 40 W lamps together.



DESIGN REQUIREMENTS

- Four lamps operate together from PLC output %Q4.1
- Four lamps: 4 x 40 W = 160 W nominal
- Nominal load current: 4 x 1.67 A = 6.67 A total
- PLC output drives the 24 VDC coil only; fit coil suppression
- Single contact must be certified for 24 VDC LED load and verified against inrush
- Four individually protected lamp branches after the common contact
- Final relay model, fuse ratings, conductors and terminals require electrical design release

0 V RETURN: dedicated lamp-supply return.

Bond/reference according to the approved cabinet design.

AL2330 IS NOT USED FOR THE FOUR MACHINE LIGHTS.

REV12 - PROJECT ENGINEERING PIPELINE

Completed architecture, remaining engineering tasks, and release gates for electrical cabinet construction.

ARCHITECTURE DEFINED | **DETAILED DESIGN + VALIDATION + RELEASE STILL REQUIRED**

Architecture	Device allocation	Detailed wiring	Safety validation	PLC / HMI build	Cabinet release	Build + FAT
COMPLETE	COMPLETE	IN PROGRESS	REQUIRED	REQUIRED	HOLD	AFTER RELEASE

COMPLETED / DEFINED

System architecture

PLC, SCALANCE XB116, IFM AL1403 masters
and field-device topology documented.

IO-Link port maps

AL102 / AL104 physical ports shown with
device allocation and connection path.

Encoder architecture

IFM RO3101 assigned to Siemens
TM Count 2x24V, 6ES7550-1AA01-0AB0.

Pilz safety architecture

PNOZmulti 2 modules, PSEN door switches
and PROFINET diagnostic boundary documented.

Stack light

Leuze TL100 command path defined through
AL102 X8 and IFM AL2330 X1 outputs.

Machine work lights

Four 24 VDC / 40 W lights switched together
by one PLC-commanded relay / contactor.

REMAINING ENGINEERING WORK

1 M104 hood fan

Confirm motor nameplate and starting method.
Select and size contactor / VFD and protection.

2 Machine lights

Select exact 24 VDC relay / contactor.
Verify LED inrush, PSU, fuses and conductors.

3 Sensors + IO-Link

Confirm pressure-sensor model and pinout.
Approve LMT100 SIO / IO-Link parameter set.

4 Pendant / JOG

Issue final pinout, contact allocation and
verified PLC addresses from product documentation.

5 Pilz safety

Complete risk assessment and PNOZmulti project.
Allocate 11 doors, E-stops, STO, EDM and restart.

6 Networks + I/O

Assign SCALANCE ports, IP plan and cable IDs.
Close AL102 / AL104 spare or planned allocations.

CABINET CONSTRUCTION RELEASE GATE

Release only after approved schematics, final BOM, cable / terminal schedule, load and protection calculations, safety validation, PLC / HMI I/O review, and issued-for-construction sign-off.